Homework 2

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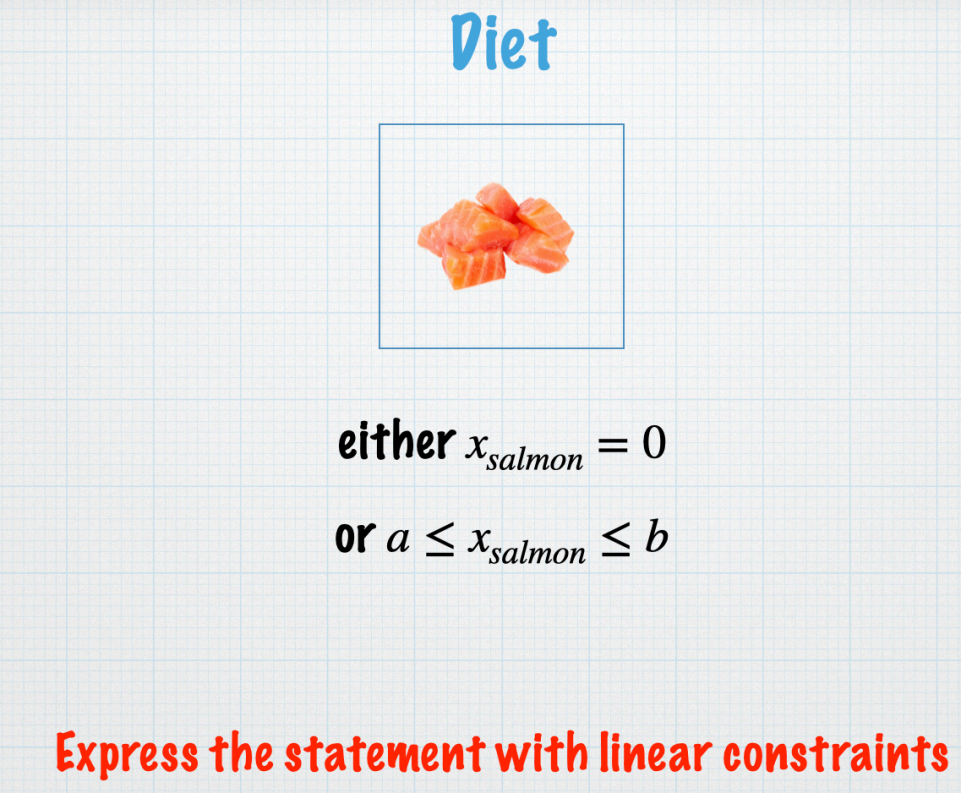

To express said statement in linear constraint we must introduce a new variable that can express the choice of picking/not picking the salmon:

$$\text{let binary variable } y \in \{0, 1\} \text{ so that:}$$

$$y_{\text{salmonChosen}} = \begin{cases} 0 & \text{if } x_{\text{salmon}} = 0 \\ 1 & \text{if } x_{\text{salmon}} \geq 1 \end{cases}$$

We can then bond together the variable of quantity and the variable of choice by writing:

$$x_{\text{salmon}} \geq a \cdot y_{\text{salmonChosen}}$$

$$x_{\text{salmon}} \leq b \cdot y_{\text{salmonChosen}}$$

This way, if y = 0 then x must be 0, if y = 1 then x must be between a and b.